

Programming 2

ECTS

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Prerequisites

Completed the 1st semester Software Engineering course "Programming 1" (formerly "Software Development with Java 1") or a similar course.

Main purpose

The purpose is to qualify the student to understand and master the concepts and techniques of object-oriented system development and programming, including Client/Server programming and basic understanding of computer networks.

Knowledge

The student must be able to explain:

- system architecture, including client/server structure and concurrent programming
- key principles of object-oriented system development, including selected design patterns and software testing methods
- basic network principles relevant to client/server communication, including layering, addressing, name systems and factors that affect network latency.

Skills

The student must be able to:

- Apply design patterns and develop object-oriented code in Java
- Develop thread-safe classes and multi-threaded programs
- Perform unit tests and system tests using relevant test techniques
- Develop client/server systems using sockets in Java
- Analyse network connections in socket-based client/server applications using packet sniffer software.

Competences

The student must be able to:

- Develop flexible and maintainable Java software using relevant architecture patterns
- Plan and implement test strategies for object-oriented systems
- Develop complete client/server solutions with self-defined application protocols
- Incorporate network considerations into the design and implementation of socket-based client/server applications.

Topics

Java, client/server systems, system architecture, design patterns, computer networks, internet protocols

Teaching methods and study activities

The semester has 96 classroom lessons. These lessons will consist of teacher presentations and exercises. The course will include a number of course assignments.

Exam

Exam prerequisites:

1. Attendance ($\geq 75\%$)
2. Three or four course assignments handed in before deadline.
3. Passed a written test in network theory. The test covers knowledge objective on network layers, addressing, DNS and network latency - topics that are not included in the oral exam.
4. For Software Engineering students only: Completed or achieved credit transfer for WS2, Programming Workshop 2. If the exam prerequisites are not met, the student must complete a written assignment in WISEflow to qualify for the re-exam. This assignment will be scheduled after the ordinary exam.

Type of exam:

Individual 20 minutes oral examination without preparation.

The student will draw from a pool of previously known questions.

The student will explain concepts and theories from the course, preferably using the course work as reference. External assessment.

Allowed tools:

All

Re-exam:

Conducted as the ordinary exam.

Grading criteria

Grading based on the Danish 7-point scale.

Responsible

Troels Mortensen (trmo)

Valid from

01-02-2026

Course type

Compulsory for Software Engineering, 2. semester